HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

ENS221: Environmental Science

Mid-term Examination

Semester: Spring 2023
Time: 1 Hour 30 Minutes
Total Marks: 30

Date: 19/ 06/ 23

Answer any **three** of the following questions: $3 \times 10 = 30$

1. a) What do you understand by the term 'Environment'? Explain. 2

   b) Environmental Science is a multi-subjective approach. Elucidate it. 4

   c) Sketch the vertical stricture of atmosphere and elucidate stratosphere and thermosphere of it. 4

2. a) The photosynthesis process: $CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$. It is an example of recycle system of nature. Justify it. 2.5

   b) Give a description about the greenhouse mechanism in your own words. 4.5

   c) Write the sources of main three greenhouse gases. 3

3. a) Define sustainable development. Write the evolution of sustainable development from 1958 to 2016. 6

   b) 'Sustainable development protects the natural resources'. Explain. 4

4. a) Write an argument on greenhouse gases emissions between developing and developed countries. 5

   b) Elaborate the term recycle, reuse, reduce, rethink and reject in the context of waste management. 5

5. a) Define Coastal Region with figure. 2

   b) Write a comparison between global warming and climate change with an example. 3

   c) What do you think about the main idea of sea level rise? Mention the effects of storm surge in the coastal region. 5

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 225: Digital Logic Design

Batch: 21st | Semester: Spring 2023
Time: 1 hour 30 minutes
Date: 21/06/2023 | **Mid-Term Examination** | Total Marks: 30

Answer any **three** questions. **Marks: 10 x 3 = 30**

1. (a) What is a Computer Memory? Why Computer Memory needed? 03
   (b) Briefly Explain the types of Computer Memory. 03
   (c) How SRAM and DRAM works? 04

2. (a) What are the differences between RAM and ROM? 03
   (b) Why control is mandatory for a system? Describe control and data-processor interaction. 03
   (c) How many methods needed for control organization briefly explain them. 04

3. (a) Compare PLA vs Control Memory. 05
   (b) Compare PLA vs Sequence Register and Decoder. 05

primary

4. (a)

TABLE 9-4 Function table for the ALU of Fig. 9-13

Selection				Output	Function
s_2	s_1	s_0	C_{in}		
0	0	0	0	$F = A$	Transfer A
0	0	0	1	$F = A + 1$	Increment A
0	0	1	0	$F = A + B$	Addition
0	0	1	1	$F = A + B + 1$	Add with carry
0	1	0	0	$F = A - B - 1$	Subtract with borrow
0	1	0	1	$F = A - B$	Subtraction
0	1	1	0	$F = A - 1$	Decrement A
0	1	1	1	$F = A$	Transfer A
1	0	0	X	$F = A \vee B$	OR
1	0	1	X	$F = A \oplus B$	XOR
1	1	0	X	$F = A \wedge B$	AND
1	1	1	X	$F = \bar{A}$	Complement A

Develop a design algorithm for Addition of two binary number using Hardwired control method for the given scenario.

5 (a) Multiply of two fixed-point binary numbers using PLA control method. 10

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 21st
Date: 05/07/2023

Mid-Term Examination

Semester: Spring 2023
Time:1 Hour 30 Minutes
Total Marks : 30

Answer any **five** questions: $5 \times 6 = 30$

1. (a) Explain the components of data communication. 3
   (b) Define three necessary criteria for an effective and efficient network. 3

2. (a) Describe the phases in periodic analog signals. 3
   (b) Calculate the corresponding periods for the frequencies listed below: 3
   I. 24Hz 41.6 ms
   II. 8MHz 0.125 μs
   III. 140KHz 7.14 μs

3. (a) Explain the modulation of a digital signal for transmission on a band pass channel. 3
   (b) Name the three types of transmission impairment. Explain. 3

4. (a) Explain bandwidth-delay product. 3
   (b) Define Bandwidth, Throughput and Latency. 3

5. (a) Write the summary of line coding schemes. 3
   (b) Distinguish between signal element and data element. 3

6. (a) Describe three different sampling methods for pulse code modulation. 3
   (b) Draw a table for 4B/5B mapping codes. 3

7. For each of the following four networks, explain the consequences if a connection fail. 6
   I. Five devices arranged in a mesh topology
   II. Five devices arranged in a star topology (not counting the hub)
   III. Five devices arranged in a bus topology
   IV. Five devices arranged in a ring topology

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming in Java

Batch: 21st / Semester: Spring 2023

Date: 08/07/2023 **Mid-Term Examination** Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions of the followings as Alphabetical order. 5 x 6 = 30

1. (a) Explain **encapsulation** of Object Oriented Programming (OOP) with appropriate example. How encapsulation ensures the data security of class members? 3

   (b) Write a Java program that accepts input as a number which represents a temperature on the Fahrenheit scale and then computes and prints its equivalent Celsius value in decimal form. Use the conversion formula $C = 5(F - 32)/9$. 3

2. (a) Rewrite the following code with **do-while** loops only. 2

(5*(F-32))/9

```
for (i=0;i<=5;i++) {
        for (j=1;j<=i;j++)
                System.out.printf("%i",i);
        System.out.println();
}
```

   (b) *Consider the following main method:* 4

```
public class TestMain {
   public static void main (String [] args) {
      int a = testf ("sum", 1,2,4,5,6); // a = 18
      int b = testf ("sum", 1,3,6); // b = 10
      int c = testf ("mult", 1, 2, 4, 5, 6); // c = 240
      int d = testf ("mult", 1,3,6); // d = 18
   }
}
```

Write the Java method **testf** in the appropriate place. You can write one or multiple **testf** method(s).

3. (a) Define instance variable and methods in Java. 2

   (b) Explain the relationship among Java Development Toolkit (JDK), Java Virtual Machine (JVM) and Java Runtime Engine (JRE). 4

4. (a) Differentiate between **equals()** and **equalsIgnoreCase()** methods with appropriate example. 2

   (b) Write a program in Java to reverse a given string without using built-in methods and check whether given string is Palindrome or not. 4

(a) What is default constructor and parameterized constructor? Briefly explain with example. 3

(b) "A method's declared return type must match the type of the value used in the parameter list." (**True / False**) 3
Justify your answer with example.

6. (a) Create a class named **Horse** that contains data fields for the **name**, **color**, and **birth year**. Include get and set methods for these fields. Next, create a subclass named **RaceHorse**, which contains an additional field that holds the number of races in which the horse has competed and additional methods to get and set the new field. Write a Java program that demonstrates using objects of each class. 2

(b) Write a method that takes an array of integers as an argument and returns a value based on the sums of the even and odd numbers in the array. Let X = the sum of the odd numbers in the array and let Y = the sum of the even numbers. The method should return X – Y. And check the return value is Prime or not. 4

7. (a) What would be the output of the following code snippet if variable **a= -10**? 2

a= -10

```
if(a<=0)
{
  if(a==0)
  {
   System.out.println("1 ");
  }
  else
  {
   System.out.println("2 ");
  }
}
System.out.println("3 ");
```

2 3

(b) Consider a **tollbooth** at a bridge. Cars passing by the booth are expected to pay a 50-cent toll. Mostly they do, but sometimes a car goes by without paying. The tollbooth keeps track of the number of cars that have gone by, and of the total amount of money collected. 4
Model this tollbooth with a class called **TollBooth**. The three data items are a list of string to hold the registration numbers of the cars passing through the bridge, a type unsigned int to hold the total number of cars, and a type double to hold the total amount of money collected. A constructor initializes both of these to 0 (zero). A member function called **payingCar()** adds the car registration number in the list, increments the car total and adds 0.50 to the cash total. Another function, called **nopayCar()**, adds the car registration number, increments the car total but adds nothing to the cash total. Finally, a member function called **display()** displays the list of cars' registration numbers and the two totals. Make appropriate member functions const. Use proper naming convention.

ABCDEFGHIJKLMNOPQRST

HAMDARD UNIVERSITY BANGLADESH

10

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221: Computer Algorithms

Batch: 21st | Semester: Spring 2023

Time: 1 Hour 30 Minutes

Date: 24/06/2023 | **Mid-Term Examination** | Total Marks: 30

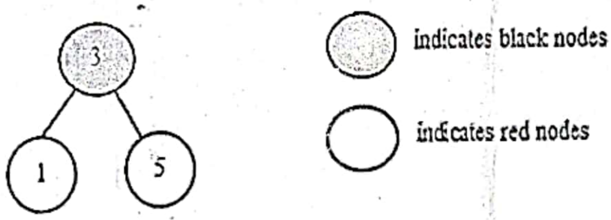
Answer any **three** questions of the followings. $3 \times 10 = 30$

1. (a) What does it mean by analysis of algorithm? Explain the types of algorithm analysis. 3

   (b) What are the reasons for worst-case analysis? 3

   (c) Insert 11, 21, 23 and 17 on the following Red Black tree: 4

   $n-k=1$



2. (a) Suppose CSE215 is a list of 12 alphabetic characters as follows: 3

   M A C R O P H Y S I C S   5

   The characters in CSE215 are to be sorted alphabetically. Use the quicksort algorithm to find position of the first character M.

   (b) Find the complexity from the following recurrence: 3

   $$T(n) = \begin{cases} 2T(n-1) - 1 & \text{if } n > 1 \\ 1 & \text{otherwise} \end{cases}$$

   $O(1)$

   (c) Let, you have two strings A= “FRIED” and B= “FIRED”. Now calculate the longest common subsequence using recursion. 4

3. (a) Let, you have 12 elements in three disjoint sets as follows: 3

   $S_1 = \{5, 8, 3, 2\}, S_2 = \{4, 7, 12, 14, 17\}$ *and* $S_3 = \{6, 10, 13\}$

   Now show all the possible representation to organize them in memory.

   (b) Multiply the following matrices using Strassen’s Matrix Multiplication formula. 3

   $A = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 4 \\ 6 & 3 \end{bmatrix}$

   (c) State the recursive Mini-Max algorithm. 4

4. (a) Consider the following instance of knapsack problem: 3

   Number of objects, $n = 6$;

   Capacity of knapsack, $m = 14$;

   Profits, $\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{10, 5, 15, 7, 21, 18\}$ *and*

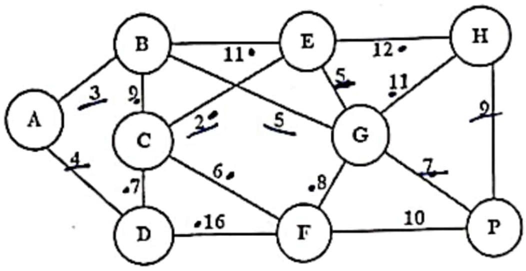
   Weights, $\{w_1, w_2, w_3, w_4, w_5, w_6\} = \{3, 2, 4, 4, 4, 5\}$.

   Calculate the optimal solution.

   57.33

(b) Define Spanning Tree. Write the general properties of Spanning Tree. 3

(c) Apply Prim's algorithm to generate the minimum Spanning Tree for the following graphs: 4



D2

HAMDARD UNIVERSITY BANGLADESH

Y

horizontal

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 21st
Date: 16/09/2023

Final Examination

Semester: Spring 2023
Time: 2 Hours
Total Marks : 40

Answer any *five* questions: $5 \times 8 = 40$

1. (a) List all the categories of multiplexing. Define FDM. — 3
   (b) Illustrate three strategies of time division multiplexing. — 5

2. (a) Explain spared spectrum and its goal. — 4
   (b) Draw and explain digital signal service (Digital hierarchy). — 4

3. (a) How does sky propagation differ from line-of-sight propagation? — 4
   (b) List the differences between omnidirectional waves and unidirectional waves. — 4

4. (a) Draw a table for categories of UTP cables. — 4
   (b) Write the advantages and disadvantages of fiber optic cable. — 4

5. (a) Explain packet switching network. — 4
   (b) Describe three phases of circuit-switched network. — 4

6. (a) Distinguish between a point-to-point link and a broadcast link. — 4
   (b) Illustrate address resolution protocol packet. — 4

7. (a) Define constellation diagram and explain its role in analog transmission. — 4
   (b) Draw the constellation diagram for the followings: — 4
      i. ASK, with peak amplitude values of 1 and 3.
      ii. BPSK, with a peak amplitude value of 2.
      iii. QPSK, with a peak amplitude value of 3.
      iv. 8-QAM with two different peak amplitude values, 1 and 3, and four different phases.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

BCVEFGH

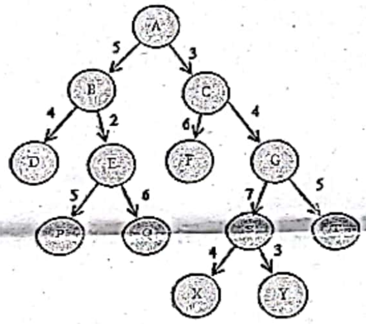
CSE221: Computer Algorithms

Batch: 21st | Semester: Spring 2023
Time: 2 Hours
Date: 11/09/2023 | Final Examination | Total Marks: 40

Answer any **four** questions of the followings. $4 \times 10 = 40$

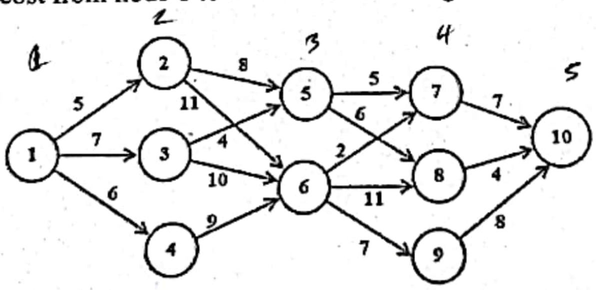
1. (a) Let you have four slots and six jobs with the following profits and deadlines respectively: 4
$\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{20, 17, 12, 10, 9, 8\}$ *and*
$\{d_1, d_2, d_3, d_4, d_5, d_6\} = \{3, 1, 4, 1, 2, 3\}$
If each job takes one unit of time to execute, find the optimal solution using Greedy Algorithm.
58

(b) Consider the tolerance level of the following network is 10. Now split the network with the help of Greedy Method. 6



2. (a) Describe the components of Dynamic Programming. Distinguish between Dynamic Programming and Recursion. 4 2

(b) Find the minimum cost from node 1 to 10 for the following multistage graph: 6



21

1 3 5 8 10

23
24

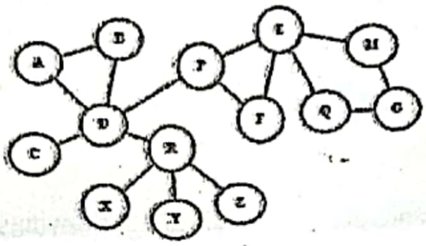
3. (a) Consider the following instance of the 0-1 knapsack problem: 5
Number of items, $n = 4$,
Capacity, $W = 8$,
Profits $\{p_1, p_2, p_3, p_4\} = \{5, 6, 7, 11\}$ *and*
Weights $\{w_1, w_2, w_3, w_4\} = \{2, 3, 3, 5\}$
Calculate the optimal solution using dynamic programming.
0010

(b) Suppose, you have two strings A= "BQDRCVEFGH" and B= "ABCVDEFGH" 5
Now calculate the longest common subsequence using the dynamic programming.

LCS=7
BCVEFGH

4. (a) Find two solutions for n-queens problem using backtracking when n = 4. 5

(b) Find the articulation points and bi-connected components of the following graph with the help of DFS Spanning tree. 5 -1



DRPE

5. (a) Apply branch-and-bound technique to solve 15-puzzle problem for the particular organizations: 5

1	2	3	4
5		7	8
9	6	10	11
13	14	15	12

(b) When a collision occurred in hashing? Make a Comparison among the Open Addressing Techniques. 5

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming in Java

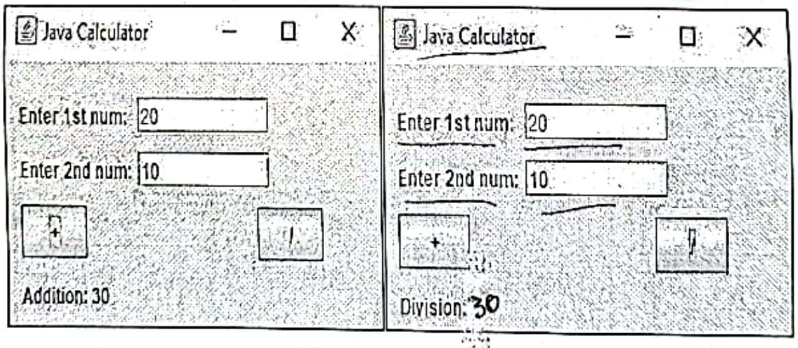
Batch: 21st | Semester: Spring 2023
Date: 18/09/2023 | **Final Examination** | Time: 2 Hours
Total Marks: 40

Answer any **four** questions of the followings. 4 x 10 = 40

1. (a) Write the names of operations in Java file. 2
   (b) What is Byte Stream and Character Stream? 2
   (c) Write a program in Java to create a file named 'write.txt' and write the text as 'We Love Our Country' in this 'write.txt' file. 6

2. (a) Which Java class is the parent class of all Exceptions? 2
   (b) What is the difference between Exceptions and Errors? 2
   (c) What are the purposes of Exceptions? Give an example of what they're used for. 6

3. (a) Define 'Thread'. Why Thread is used in Java? 2
   (b) Write a Java program to display the following interface using AWT, Swing/Applet class with ActionListener. 8



4. (a) Explain the life-cycle of Applet in Java. 4
   (b) Write a program in Java using Swing for text **"Object Oriented Programming"**, where text font name is **"Roboto"**, size 25 and in **bold** format. Also **blue** as font color and **green** as background color. 6

5. (a) Write the features of **enum** in Java. Write a program in Java to demonstrate the use of **values(), valueOf()** and **ordinal()** methods. 5
   (b) Explain the life-cycle process of Java Multithreading. 5

$\overline{\overline{A+B}}$
$\overline{A}.\overline{B}$

HAMDARD UNIVERSITY BANGLADESH

$\overline{A}B+A\overline{B}$
$10+01$
0

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE225: Digital Logic Design

Batch: 21st
Semester: Spring 2023

Date: 09/09/2023
Final Examination
Time: 02:00 Hours
Total Marks: 40

Answer any **five** questions. **Marks: 8 × 5 = 40**

1. (a) What is meant Boolean Algebra? 01
   (b) What is universal gate? Prove that, NAND gate is a universal gate. 03
   (c) Derive the truth tables of different logic gates. 04

2. (a) Define SOP and POS. 02
   (b) Draw a circuit diagram for Y = (A + BC) (B + CA). 02
   (c) There are two methods in Boolean functions: Standard form and Canonical form. Convert the expression F= X+YZ to Canonical form two terms are Sum of minterm and product of Maxterm. 04
   $\Sigma(3,4,5,6,7)$
   $\Pi(0,1,2)$

3. (a) What are the differences between Flip-flop and Latch. 02
   (b) What would be the elaborate form of Y if F (A, B, C) = ΠM (0,1,2,5)? 02
   (c) Convert the Sum of product (SOP) expressions to Product of sum (POS) form F (x, y, z) = y'+ x.z! 04

$\overline{y}+(\overline{x+z})$
$y'+$

4. (a) What is Karnaugh Map? Write the uses of K-map. 03
   (b) Given F (W, X, Y, Z) = ∑m (0,4,5,6,12,14,15), find a minimized form using K-map. $x\overline{z}+\overline{w}\overline{y}\overline{z}+\overline{w}x\overline{y}+wxy$ 05

5. (a) Implement a full adder logic circuit using AND, XOR, and OR gate or from two half adder. 04
   (b) Simplify the following function using K-map. Enter F (A, B, C, D) = ∑ (0,1,2,9,11) + d (A, B, C, D) = ∑ (8,10,14,14). 04
   $A\overline{B}+\overline{B}\overline{D}+\overline{B}\overline{C}$

A,B,C
A',B',C'
$A+B+C$
$(\overline{A+B+C})$
$\overline{A}'.\overline{B}.\overline{C}$
$(\overline{ABC})$
$\overline{A}+\overline{B}+\overline{C}$

6. (a) Write are the differences between combinational and sequential circuit. 03
   (b) If you are asked to design a 4:1 multiplexer, how many select inputs will you use? Why? Draw the circuit diagram of such a multiplexer. 05

7. (a) K-map is mainly used for the simplification of complex Boolean expressions. Draw a four variable K-map with proper designation. 02
   (b) Prove De-morgan's theorems for three variables. 03
   (c) Sequential circuits are commonly used in digital systems to implement state machines, timers, counters, and memory elements. Draw the truth table and a circuit diagram of a S-R Flip-flop. 03

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

ENS221: Environmental Science

Semester: Spring 2023

Final Examination

Time: 2 Hours

Date: 05/ 09/ 23

Total Marks: 40

Answer any **four (4)** of the following questions: $10 \times 4 = 40$

1. a) Define Environmental Science. What are the major components of environment? Describe each of them — 4.5

   b) What do you understand by the term sustainable development. Explain the sustainable development by the 3E concept. — 5.5

2. a) "Plastic waste is a big threat for Bangladesh." Explain. — 5

   b) Mention the benifits of rooftop farming in the urban areas of Bangladesh. — 5

3. a) Elucidate the drought mechanism in Bangladesh and adaptation processes of it. — 4 water

   b) Sketch a flow chart of flash flood and damage. Provide some suggestions for recovering from the flash flood. — 6 supply

4. a) Vulnerability = Number of Disasters × 1/ Adaptive capacity. Explain this equation in your own words. — 5

   b) Elaborate the mitigation and low carbon development in the context of Climate Change. Mention three programs of it. — 5

5. a) Write the six pillars of Bangladesh Climate Change Strategy and Action Plan (BCCSAP). — 3.5

   b) The food security, social protection and health is the most important pillar among six pillars of the BCCSAP. What do you think? Justify it. State the three programs of it. — 6.5

6. a) What do you think about the main idea of Green Tourism? Illustrate the elements of the Green Tourism. — 7

   b) Make a list of threats due to thunderstorm in Bangladesh. — 3